

STRUCTURAL ANALYSIS

Self
confidence

Achieve
Daily Target

Multiple
revision

To The Point-ByDhyanPal(ESE'17AIR-179,GATE'18AIR-93,GATE'16AIR-145)



Determinacy :-

Degree of static Indeterminacy (D_s)

$$D_s = D_{se} + D_{si}$$

external Indeterminacy

Internal Indeterminacy

$$\therefore D_{si} = D_s - D_{se}$$

(2)

~~only f_y, M~~

only f_y, M

hence only 1 restrain (f_x) is added to make it fix.

(3)



rxn f_x, m

hence only 1 restrain added f_y to make it fix

Type-1 frame (D_s)

(2D)

(3D)

$$D_s = 3C - R'$$

$$D_s = 6C - R'$$

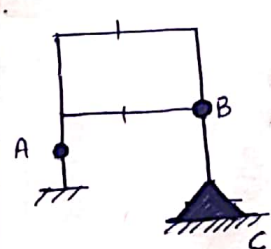
if Hinge then $R' = m - 1$

if Hinge $R' = 3(m - 1)$

$C \rightarrow$ min no. of cut required to open truss structure
 $R' \rightarrow$ Restrain added to make fix joint

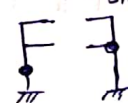
Type-1 Practice Problem :-

(2D)



$$D_s = 3C - R'$$

$C = 2$ { 2 cut req. to make truss.



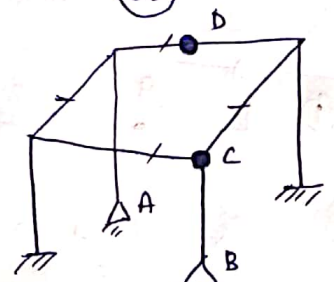
$$R' = A + B + C = 4$$

(m-1) 2-1 (1)

m-1 3-1 (2)

already hinge hence only 1 restrain M added to make it fix (1)

(3D)



$$D_s = 6C - R'$$

$C = 4$

$$R' = A + B + C + D$$

(3) (3) 3(m-1) 3(m-1)
 Restrain added to make fix (6) (3)

$$Ans \Rightarrow 6 \times 4 - 15 = 24 - 15 = 9$$

How to get D_{se} :-

$$D_{se} = R - S$$

no. of available equilibrium equation

external no. of reaction generated

R table

S table

	2D	3D
Roller	only f_y hence $r=1$	$R=1$
Hinge	$r=2$ f_x, f_y	$r=3$ f_x, f_y, f_z
Fix	$R=3$ f_x, f_y, M_z	$R=6$ $f_x, f_y, f_z, m_x, m_y, m_z$

2D str

3D str

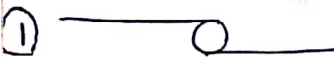
$S=3$

$S=6$

f_x
 f_y
 M_z

f_x m_x
 f_y m_y
 f_z m_z

Special :-



only vertical rxn (f_y)

hence to make it fix we have to add 2 restrain f_x, M

$$Ans: D_s = 3 \times 2 - 4 = 2$$

Type-2

Trauss case :-



$$D_s = (m+r) - 2j$$

$$D_s = (m+r) - 3j$$

$m \rightarrow$ no. of truss member
 $r \rightarrow$ no. of external reactions
 $j \rightarrow$ no. of truss joints

basically $(m+r) \rightarrow$ no. of rxn available total

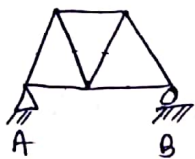
$2j$ or $3j \rightarrow$ no. of eqn available

$2j \rightarrow (2D) \rightarrow f_x, f_y = 0$

$3j \rightarrow (3D) \rightarrow f_x, f_y, f_z = 0$

Type-2

In exam 2D truss is asked.



$$D_s = (m+r) - 2j$$

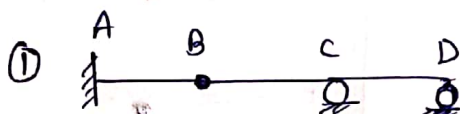
$$m=7, r=3, j=5$$

$$D_s = 7+3-2 \times 5 = 0$$

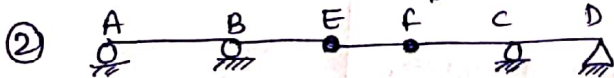
Type-3

Beam case :-

$$D_s = \text{Support reaction removed to make cantilever} - \text{Restrain added to make fix} \quad (R')$$

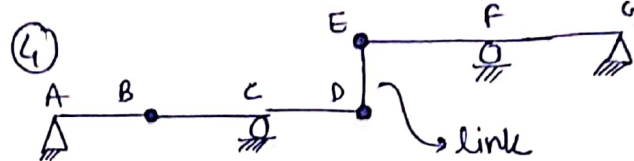


$$D_s = (1+1) - (2-1) = 1$$



$$D_s = (1+1+2) - \left(\frac{m-1}{E} + \frac{m-1}{F} + 2 \right) = 0$$

$$D_s = (3) - \left(\frac{m-1}{B} + 2 \right) = 0$$



$$D_s = (C+F+G) - (A+B+D+E)$$

$$D_s \Rightarrow 0$$

KI

Cinematic Indeterminacy :: no. of unknown joint displacements like displacement (Δ) & rotation (θ)

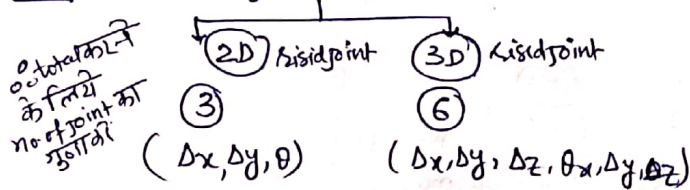
Trick for frame (KI) :- चौर

हलवा

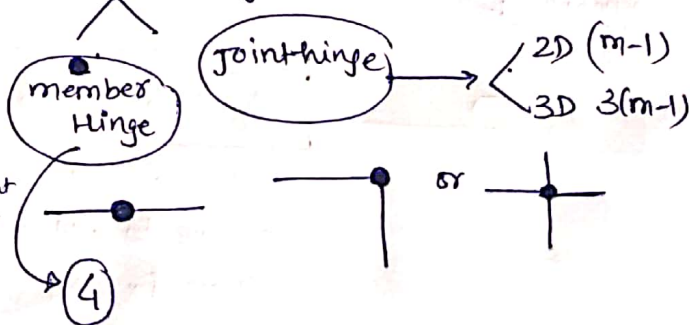
DHIRE

$$KI = (D + H) - (I + r_e) \text{ or } D+H-I-r_e$$

note:- $D \rightarrow$ Degree of Freedom (DOF)



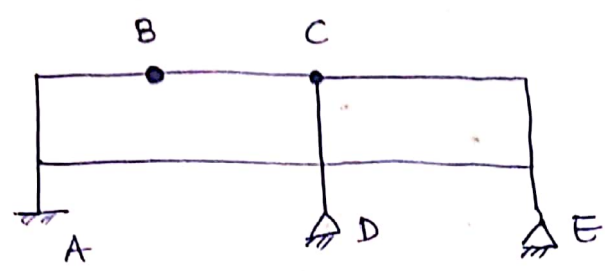
$H \rightarrow$ Hinge DOF (degree of freedom)



$I \rightarrow$ Inextensible member

$r_e \rightarrow$ no. of external rxn.

Practice Frame KI :-



$$KI = (D + H) - I - r_e$$

$$D = 3 \times \text{no of joints} = 27$$

$$H = B + C = 6$$

member hinges $\downarrow$ 4

joint-hinge $\downarrow$ (m-1) & -1 = 2

$$I = \text{Inextensible member} = 0$$

[otherwise restriction]

$$r_e = A + D + E = 7$$

3 2 2

$$KI = (27 + 6) - 0 - 7 = 26$$

Stability of structure :-

① External stability :-

Structure sufficiently restraint so that rigid body movement does not happen.

condition :-

① Three reactions in plane structure neither be concurrent & nor parallel

{ अन्यथा unstable हो जायेगा }

② In space structure

• rxn should be { non parallel, non concurrent, non coplanar }

② Internal stability :-

• when part of structure moves appreciably with respect to other part.

Truss (KI)

2D

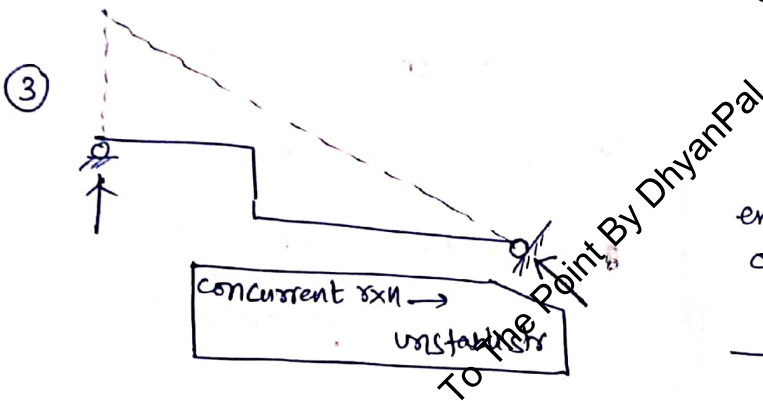
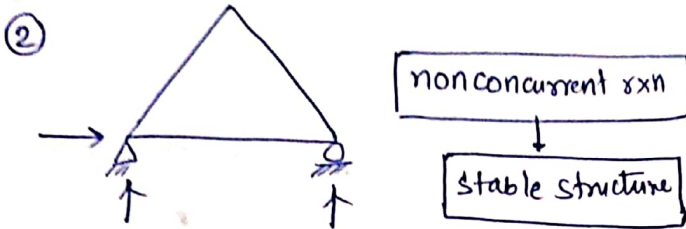
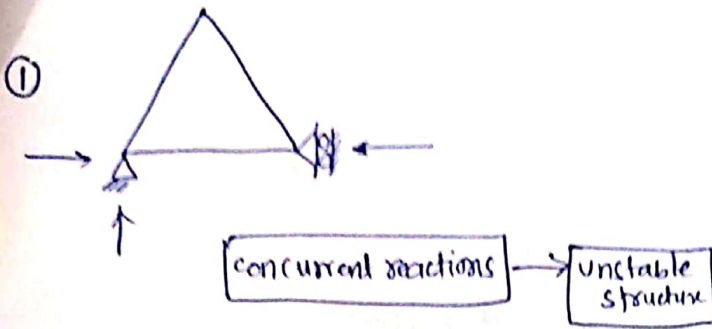
$$KI = 2j - r_e$$

3D

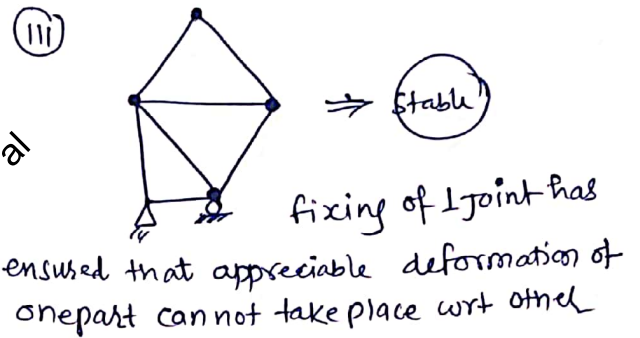
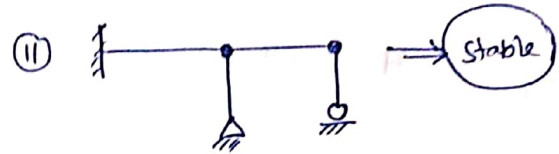
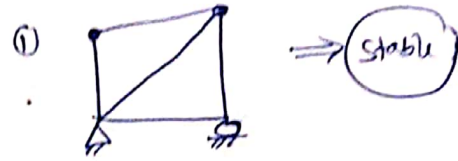
$$KI = 3j - r_e$$

Ex. :-

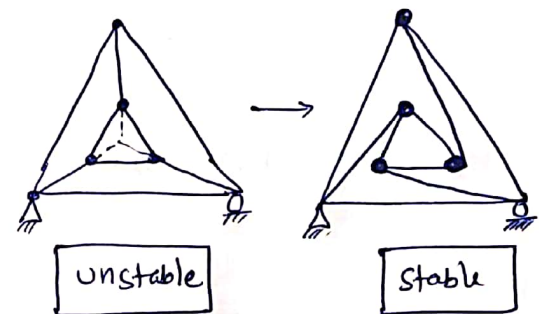
④



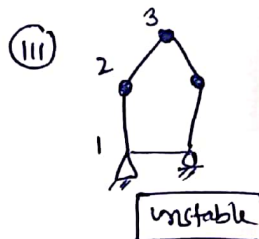
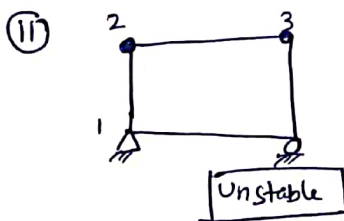
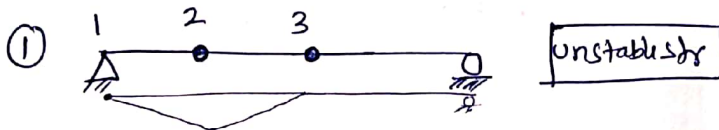
like in previous (ii) example
if 1 & 3 connected by member
then it will be stable structure.



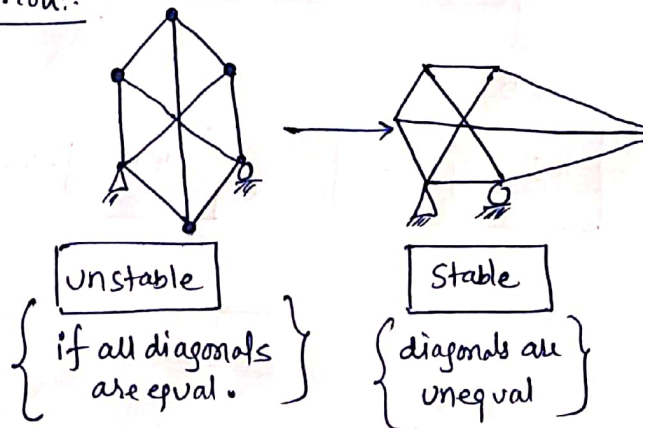
note:- Triangle within Triangle



note:- 3 hinge in continuation → mechanism form
hence unstable structure



note:-



Special case :- but if there is bracing
or additional members. then it will
be stable.

Truss :-

assumption

- pin joint
- loading at joints
- self wt neglected

Truss deflection

Angle
weight
method

joint
displacement
method.

williat
moir
method

Graphical
method

$N \rightarrow$ due to external load
type force

n type force $\rightarrow$ due to unit load } apply
unit load in that direction
in which deflection is needed

total deflection = $\frac{nNL}{AE} + nL\alpha\Delta T + n\delta L$

$\downarrow$ due to force $\downarrow$ due to temp. change $\downarrow$ due to member too short or too long.

$$F = P + Kx$$

$$x = \frac{-\sum PKL/AE}{\sum K^2L/AE}$$

forms of Truss

(i) Pratt Truss :-

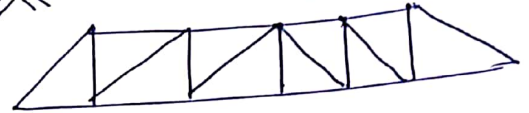
for the loads
in gravity direction.



Pratt truss

(ii) Howe Truss :-

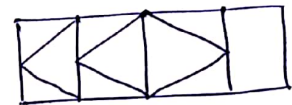
just opposite of
Pratt truss



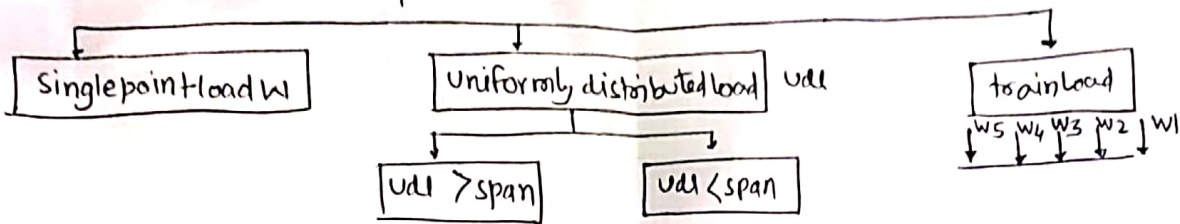
(iii) Warren Truss



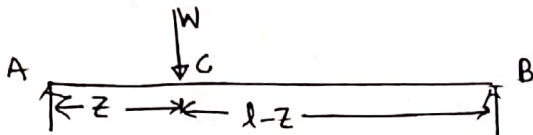
(iv) KC Truss :-



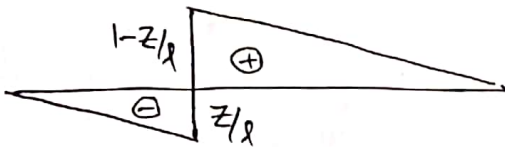
Types of moving loads on simply supported beam



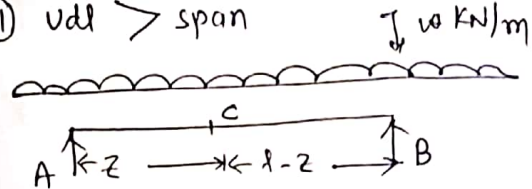
① Single point load :-



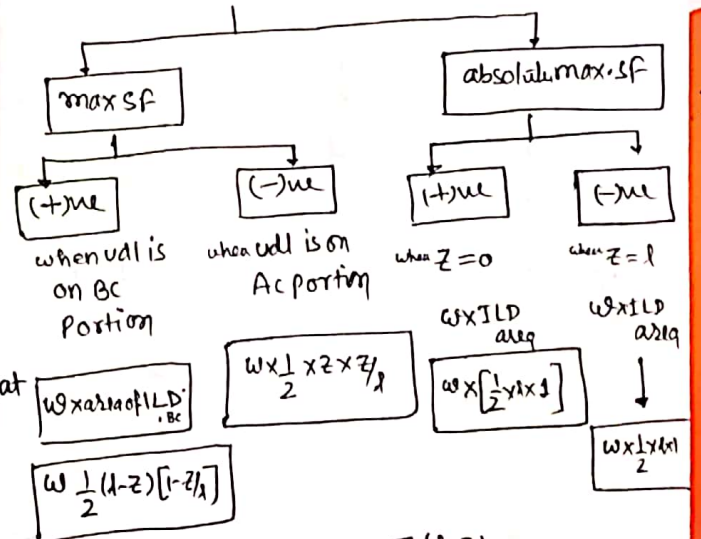
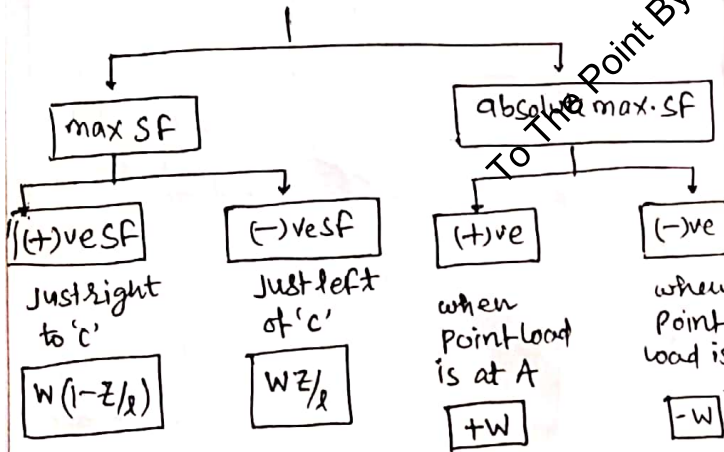
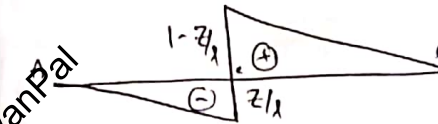
ILD for shear force (SF)



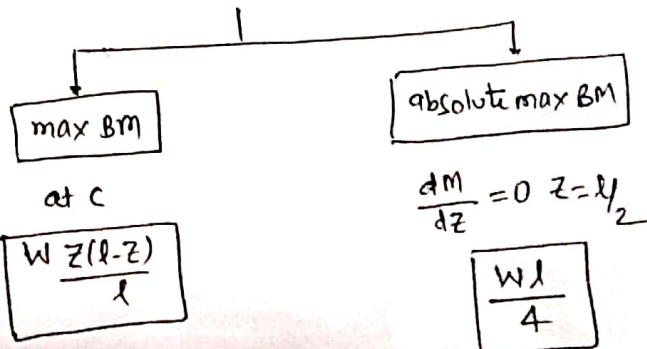
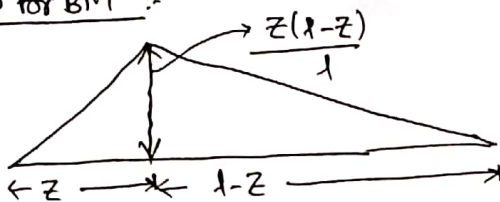
② udl > span



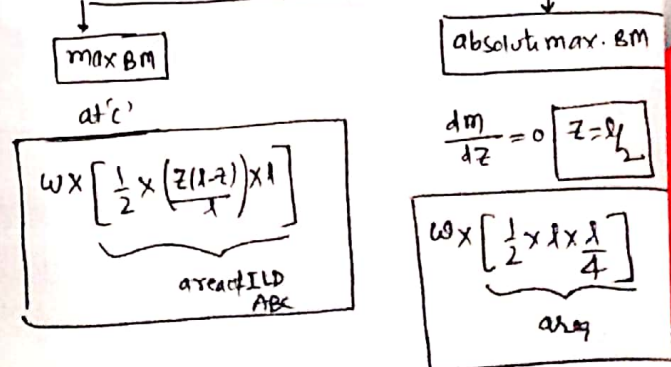
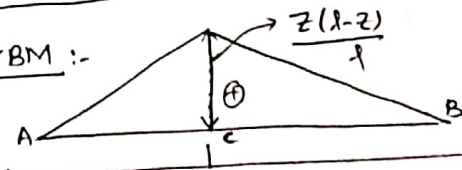
ILD for SF



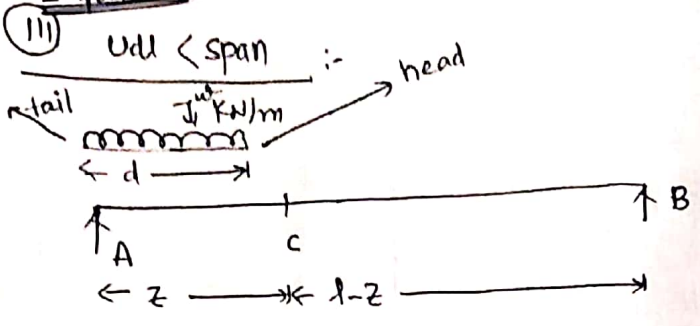
ILD for BM :-



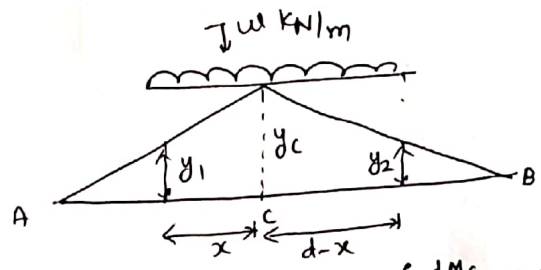
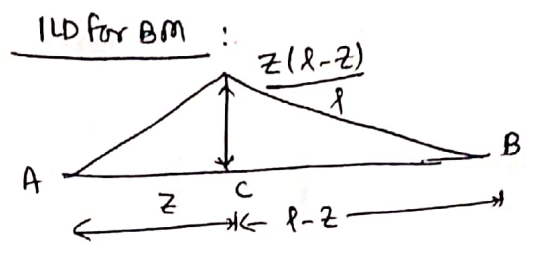
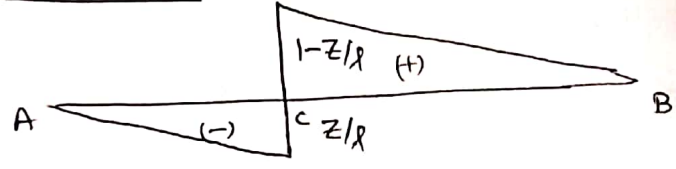
ILD for BM :-



Important



ILD for SF :-



for max. BM at C :- $\left(\frac{dM_c}{dx} = 0\right)$

$$M_c = \left[\frac{1}{2} \times x \times (y_1 + y_2) \right] w + \left[\frac{1}{2} \times (d-x) \times (y_2 + y_1) \right] w$$

$$\frac{dM_c}{dx} = 0$$

$$y_1 = y_2$$

$$\frac{z}{l} = \frac{x}{d}$$

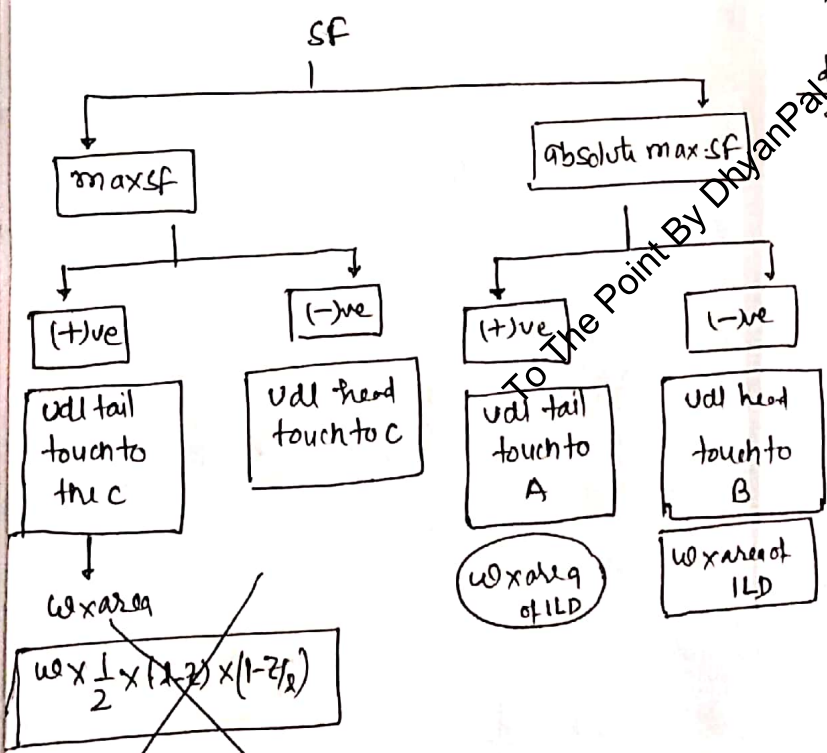
ordinates of head & tail are equal

Section divide the load in the same ratio as of span.

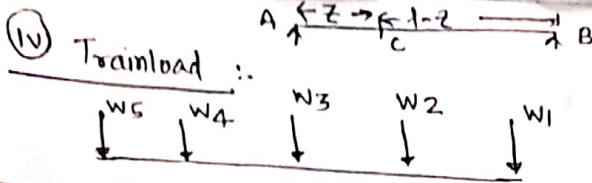
V.M.P

absolute max. BM :-

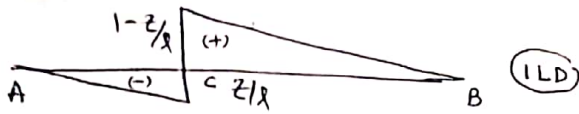
when $y_c \rightarrow \max$ then $z = \frac{l}{2} \sqrt{\frac{d}{l}}$



note :- whether it covers whole CB or not we don't know.



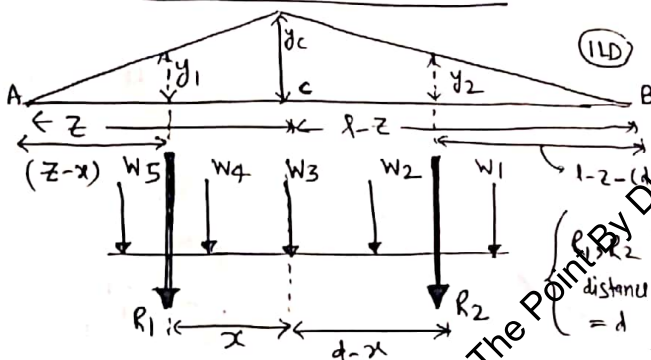
Case-1 :- max. SF at any section 'c' :-



max. (-ve) SF at c :- put W_1 at c, rest left of c] do trials
 put W_2 at c, _____]

max. (+ve) SF at c :- put W_5 at c, rest right of c] do trials
 put W_4 _____]

Case-2 max. BM at any section 'c'



let W_3 at 'c' $R_1 \rightarrow$ Resultant of loads left side of c
 $R_2 \rightarrow$ Resultant of loads right side of c

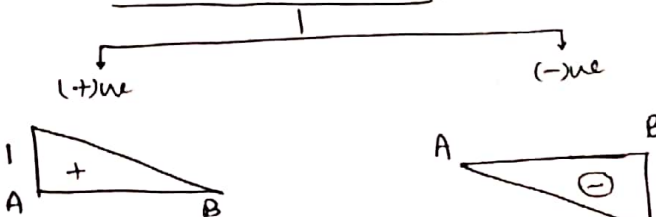
BM at c :- $M_c = R_1 y_1 + R_2 y_2$ $\begin{cases} y_1 = y_c \times \frac{z-x}{z} \\ y_2 = y_c \times \frac{1-z-(d-x)}{(1-z)} \end{cases}$

For BM at c $\frac{dM_c}{dx} = 0$

$$\frac{R_1}{z} = \frac{R_2}{1-z}$$

Conclusion :- average of loads on left and right side of section should be same
 * average of load = $\frac{\text{Resultant load (sum)}}{\text{distance}}$

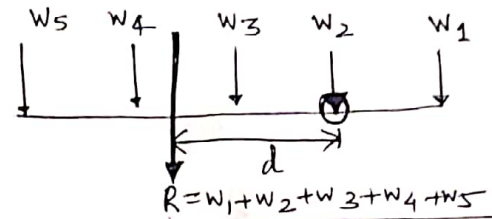
Case-3 :- absolute max. shear force :-



one of the loads should be at A, rest do the trials.

one of the load should be at B, rest do the trials

Case-4 :- max. moment under a given load (suppose W_2)



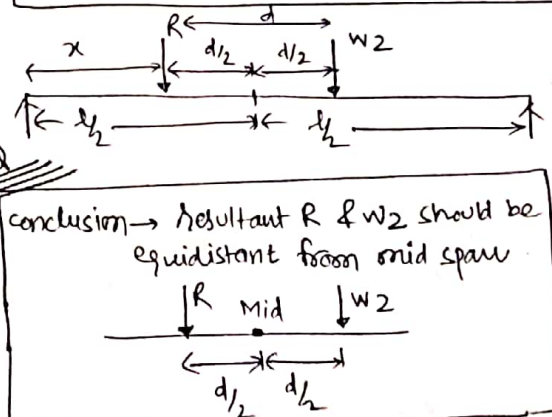
$R \rightarrow$ resultant of all loads which is 'd' distance from W_2 and 'x' distance from A.

$$\frac{R(1-x)}{1} = R_A \quad \frac{R x}{1}$$

$$\text{BM under } W_2 = R_A(x+d) - R d$$

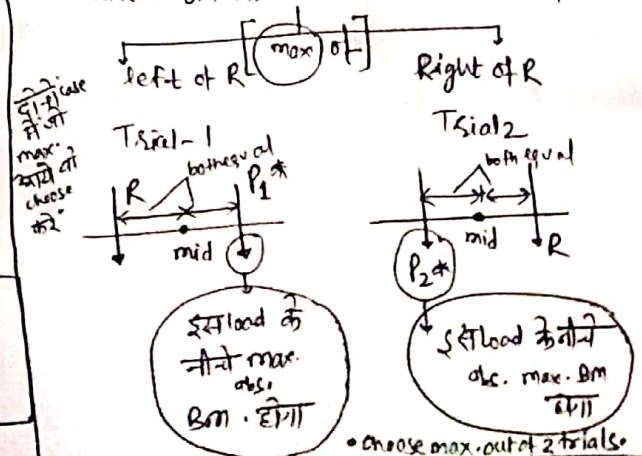
for max. $\frac{d(M \text{ under } W_2)}{dx} = 0$ $\boxed{x = \frac{l}{2} - \frac{d}{2}}$

hence W_2 distance from A $= x + d = \frac{l}{2} + \frac{d}{2}$



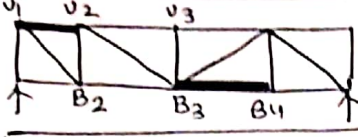
Case-5 :- absolute max. BM :-

- when heavier loads near centre
- max. moment under a wheel load



ILD for Truss member

Top chord member
or
bottom chord member



Ex. ILD for U_1, U_2 members
ILD for B_3, B_4 "

Inclined member

Ex. ILD for U_2, B_3 member

vertical member

Ex. ILD for U_2, B_2, U_3, B_3

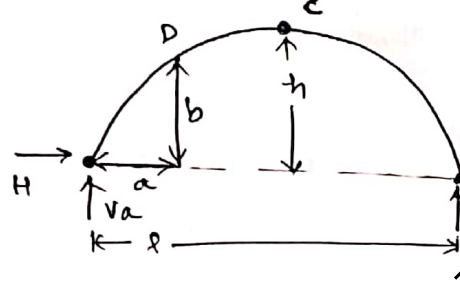
Approach :-

left panel & right of
this member

approach $\rightarrow$ same approach

Approach :- left that panel
& unit load placed in rest
of panel, first left of
that panel second, right of
that panel and then solve.

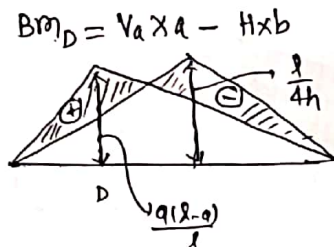
ILD for 3-hinge arch :-



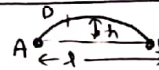
ILD of Horizontal
reaction H



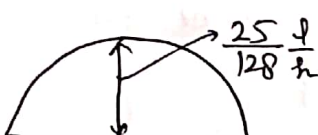
ILD for BM :-



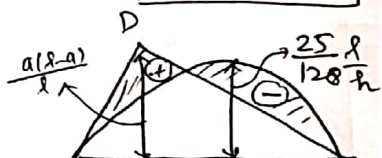
ILD for 2-hinge arch :-



ILD for H



ILD for BM



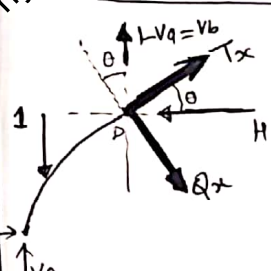
* note :-

this shaded area
depends on
values.

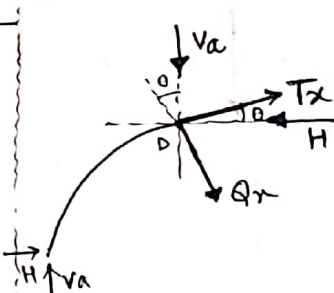
3-hinge arch :-

ILD for normal thrust & radial shear at D :-

Case-1: unit load before D



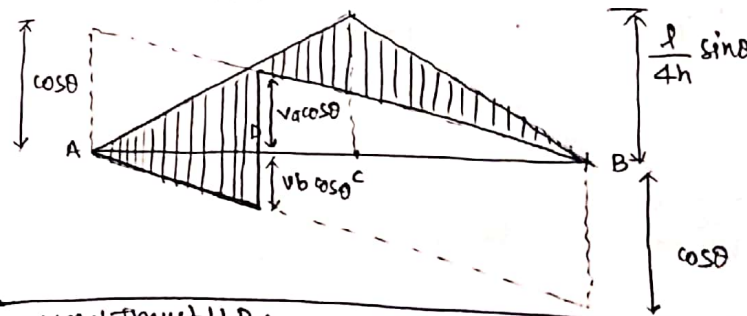
Case-2: unit load after D



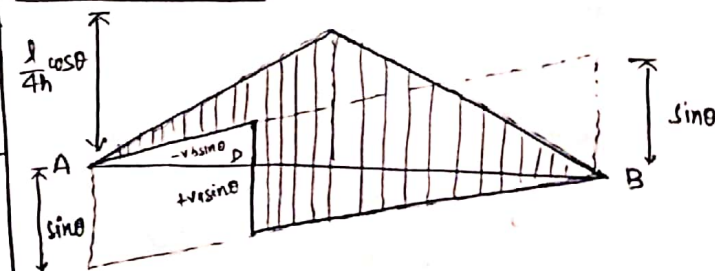
Radial shear at D $Q_x = H \sin \theta + V_b \cos \theta$ {when unit load left of D}
 $Q_x = H \sin \theta - V_a \cos \theta$ {right of D}

normal thrust at D $T_x = H \cos \theta - V_b \sin \theta$ {when unit load left of D}
 $T_x = H \cos \theta + V_a \sin \theta$ {right of D}

Radial shear ILD :-

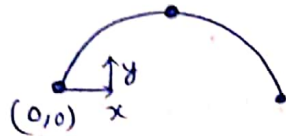


normal thrust ILD :-



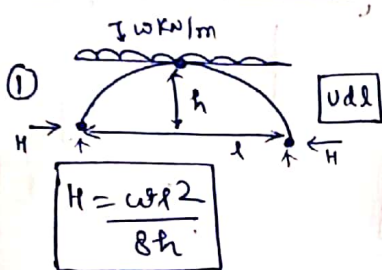
3 hinge arch

Parabolic arch

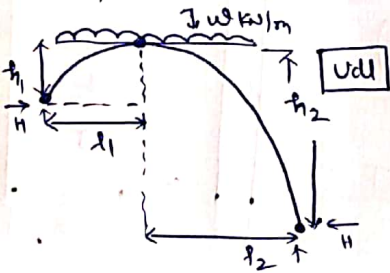


Parabolic arch

$$y = \frac{4hx(1-x)}{12}$$



② supports not at same level



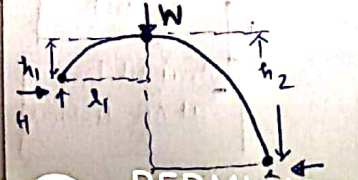
$$l_1 = \frac{l \sqrt{h_1}}{\sqrt{h_1} + \sqrt{h_2}}$$

$$l_2 = \frac{l \sqrt{h_2}}{\sqrt{h_1} + \sqrt{h_2}}$$

$$H = \frac{wl^2}{2(\sqrt{h_1} + \sqrt{h_2})^2}$$

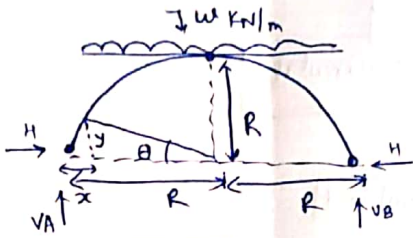
→ same type formula summit cum

③ supports not same level
 But point load at crown



$$H = \frac{Wl}{(\sqrt{h_1} + \sqrt{h_2})^2}$$

Semicircular arch

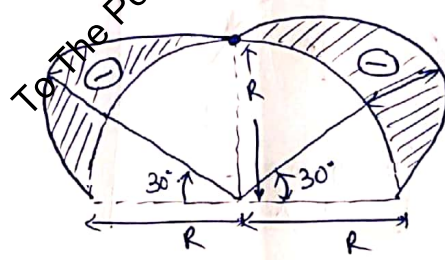


$$H = \frac{wR}{2}$$

$$BM_x = VA \cdot x - \frac{wx^2}{2} - Hy$$

x, y → components of R

$$\therefore BM_{max} = \frac{wR^2}{8} \text{ @ } \theta = 30^\circ$$



BMD

effect of temperature change



$$\Delta h = \left(-h + \frac{l^2}{4h} \right) \alpha \Delta T$$

new crown height

$$\begin{cases} \therefore H = \frac{wl^2}{8h} \\ H \propto \frac{1}{h} \\ \frac{dH}{H} = -\frac{dh}{h} \end{cases} \left\{ \begin{array}{l} \text{Parabolic} \\ \text{will } \textcircled{3} \\ \text{rise} \end{array} \right.$$

special case

3 hinge or 2 hinge arch

if fully UDL loaded in whole span

then

(i) $BM_x = 0$ (always)

(ii) Radial shear $_x = 0$ (always)

But

(ii) Normal thrust $_x \neq 0$

2 Hinge arch

$$H = \frac{\int \frac{m y ds}{EI}}{\int \frac{y^2 ds}{EI}}$$

$$I = I_0 \cos \theta$$

moment of Inertia of arch at crown
($\theta = 0 \therefore I = I_0$)

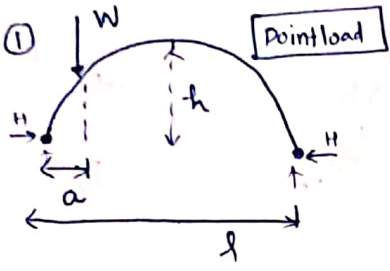
$$y = \frac{4hx(1-x)}{l^2}$$

valid

Parabolic

Semicircular

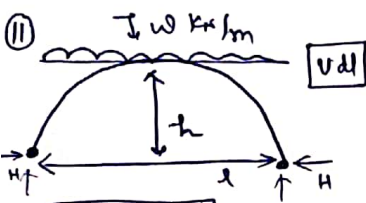
effect of temperature



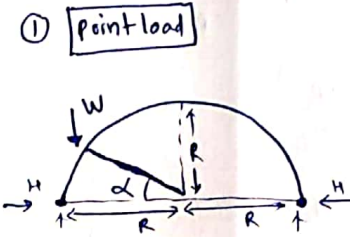
$$H = \frac{5}{8} \frac{W a (l-a)(al-a^2+l^2)}{h l^3}$$

** if $a = l/2 \Rightarrow$ point load is in mid

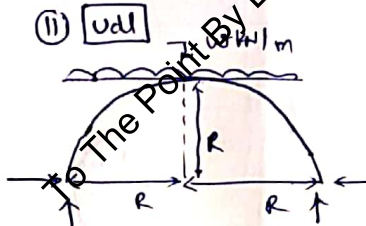
$$H = \frac{25}{128} \frac{W l}{h}$$



$$H = \frac{w l^2}{8h}$$

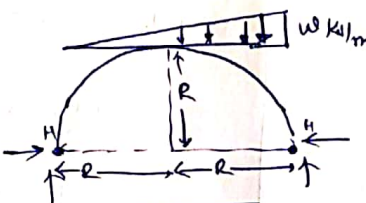


$$H = \frac{W}{\pi} \sin^2 \alpha$$



$$H = \frac{4}{3} \frac{w R}{\pi}$$

iii) Half udl or triangular loading



$$H = \frac{2}{3} \frac{w R}{\pi}$$

$$H = \frac{\int L \Delta T}{\int \frac{y^2 ds}{EI}}$$

Parabolic

Semicircular

$$H = \frac{\alpha \Delta T}{\left(\frac{8}{15} h^2 \right) / EI}$$

or

$$H = \frac{15 EI \alpha \Delta T}{8 h^2}$$

$$H = \frac{\alpha \Delta T}{\left(\frac{\pi R^2}{4} \right) / EI}$$

or

$$H = \frac{4 EI \alpha \Delta T}{\pi R^2}$$

Eddy's Theorem:



BMD at any section x & y

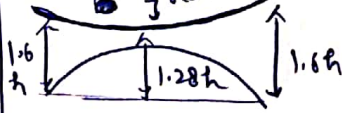
linear arch & given arch.

Trick

Reaction locus of 2 Hinge arch

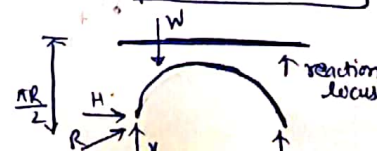
Parabolic arch

Curved (Parabolic) @
1.6h distance at ends &
1.28h distance at mid.



Semicircular arch

Straight line
at $\frac{\pi R}{2}$ distance



method of structure analysis

force method
or
flexibility method
or
compatibility method

$D_s \rightarrow$ degree of static indeterminacy
 $D_k \rightarrow$ degree of kinematic indeterminacy

Use when $D_s < D_k$

Displacement method
or
stiffness method
or
equilibrium method

Use when $D_k < D_s$

Example :-

- (i) unit load method / virtual work method By Bernoulli
- (ii) consistent deformation method
- (iii) Three moment theorem method
- (iv) column analogy method
(condition - Rigid frame with fix support)
- (v) castigliano's theorem of minimum strain energy
- (vi) elastic centre method
- (vii) maxwell - mohr equation

example :- Trick $SMS + KM$

- (i) Slope deflection method (S) $\rightarrow$ G. A. Maney method
- (ii) moment distribution method (M) $\rightarrow$ Hardy cross method
- (iii) stiffness method (S)
- (iv) kani's method
(Successive approximation, rotational coefficient)
Iterative approach for applying slope deflection method
- (v) men potential energy method.

Strain energy stored

due to axial load	due to bending	due to shear force	due to torsion
Ex. In truss	Ex. In beam, frame		
$\frac{\int P^2 dx}{2AE}$	$\frac{\int M^2 dx}{2EI}$	$\frac{\int S^2 dx}{2AG}$	$\frac{\int T^2 dx}{2GIp}$

$AE \rightarrow$ Axial rigidity
 $EI \rightarrow$ Flexural "
 $GIp \rightarrow$ torsional rigidity

$\frac{AE}{L} \rightarrow$ axial stiffness
 $\frac{EI}{L} \rightarrow$ Flexural stiffness
 $\frac{GIp}{L} \rightarrow$ Torsional stiffness

$D_s < D_k$ force method	$D_k < D_s$ Displacement method
redundant forces or reactions are taken as unknown	displacements (θ, Δ) are taken as unknown
compatibility condition is used to get or find unknown reaction/force	equilibrium equation at joints are required to find unknown displacement
no. of compatibility equation $\Rightarrow D_s$	no. of equilibrium condition / eqn = D_k

Castigliano's 1st theorem $\rightarrow$

$$\left[\frac{\partial U}{\partial \Delta} = P \right] \left[\frac{\partial U}{\partial \theta} = M \right]$$

$U \rightarrow$ total strain energy

$\Delta \rightarrow$ displacement in direction of force P

$\theta \rightarrow$ rotation ————— moment M

WIMP

Castigliano's 2nd theorem $\therefore$

$$\left[\frac{\partial U}{\partial P} = \Delta \right] \left[\frac{\partial U}{\partial M} = \theta \right]$$

Apply P to get Δ in P direction.
Apply M to get θ in that sense. (direction)

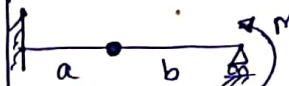
$$\text{carry over factor} = \frac{\text{carry over moment}}{\text{Applied moment}}$$



$$\text{cof} = \frac{1}{2}$$



$$\text{cof} = 0$$



$$\text{cof} = a/b$$

Betti's Law $\therefore$

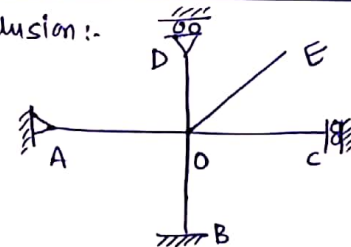


$$P_1 \Delta_1 = P_2 \Delta_2$$

Special case of Betti's law $\rightarrow$ Maxwell reciprocal theorem

$$\text{Distribution factor DF} = \frac{\text{Stiffness of member}}{\text{Sum of stiffness of all member at that joint}}$$

conclusion $\therefore$



$$M_{OA} = \left(\frac{3EI}{L} \right) \theta$$

$\because$ far end is hinge

$$M_{OB} = \left(\frac{4EI}{L} \right) \theta$$

$\because$ far end is fix

$$M_{OC} = \left(\frac{EI}{L} \right) \theta$$

$$M_{OD} = 0$$

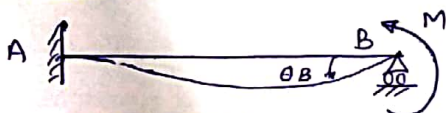
$\because$ far end is roller
 $\rightarrow$ free

moment distribution method (Hardy cross method) $\therefore$

$$\text{Stiffness} = \frac{\text{force (P)}}{\text{deflection } \Delta} = \frac{\text{moment (m)}}{\text{rotation } \theta}$$

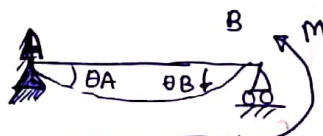
Stiffness of Beam

Stiffness of member BA when far end A is fix



$$k = \frac{4EI}{L}$$

Stiffness of member BA when far end A is hinge



$$k = \frac{3EI}{L}$$

Relative stiffness

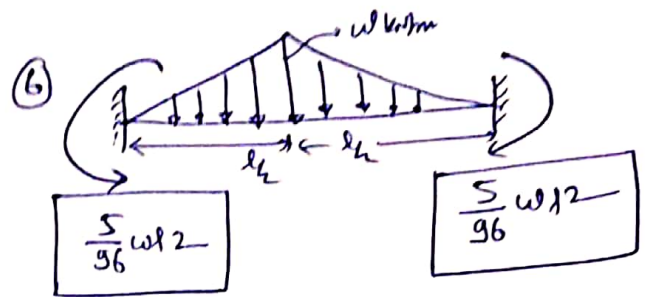
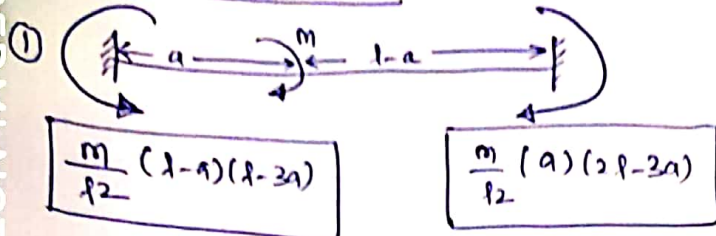
when far end is fix

$$R.S = I/L$$

when far end is hinge

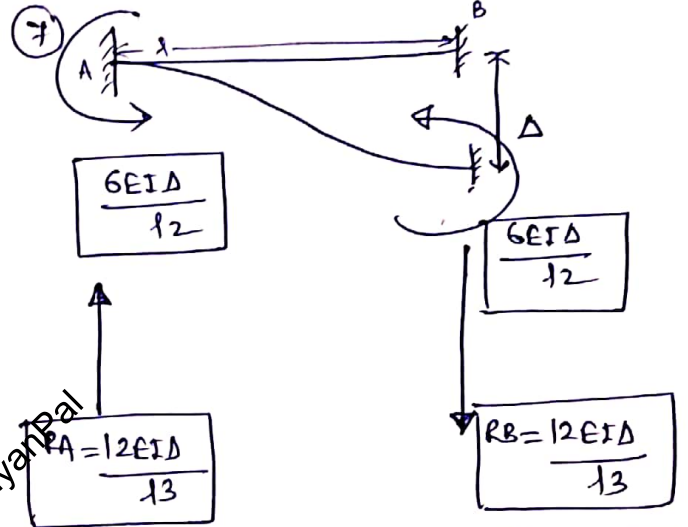
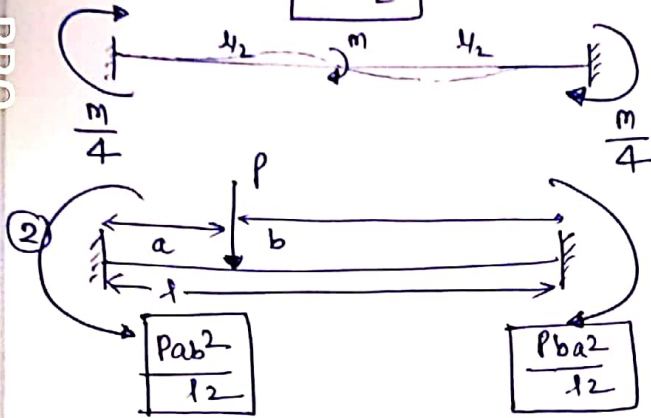
$$R.S = \frac{3I}{4L}$$

fixed end moment list :-



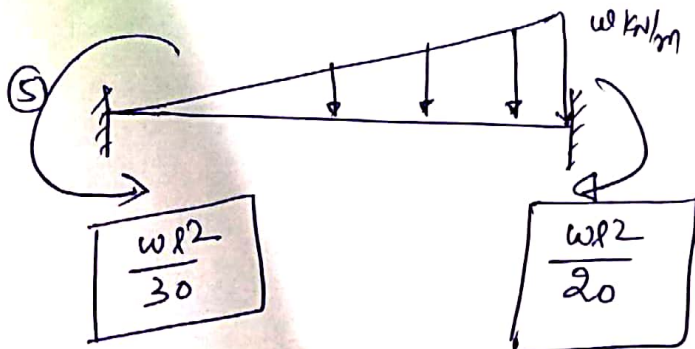
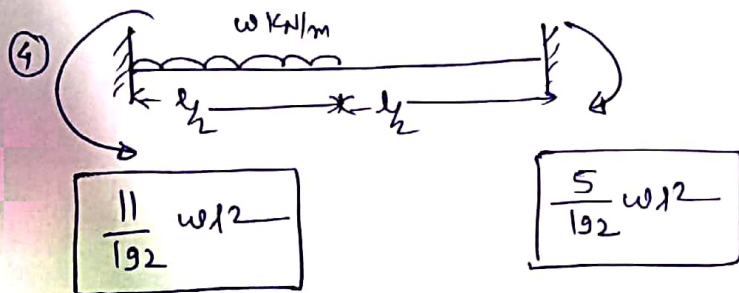
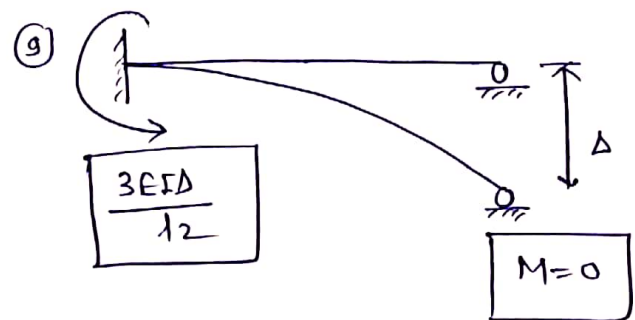
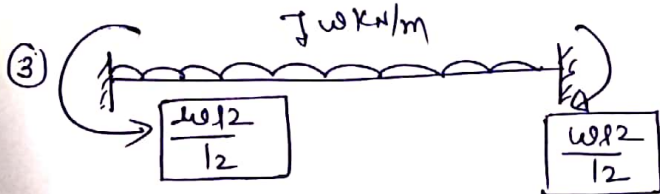
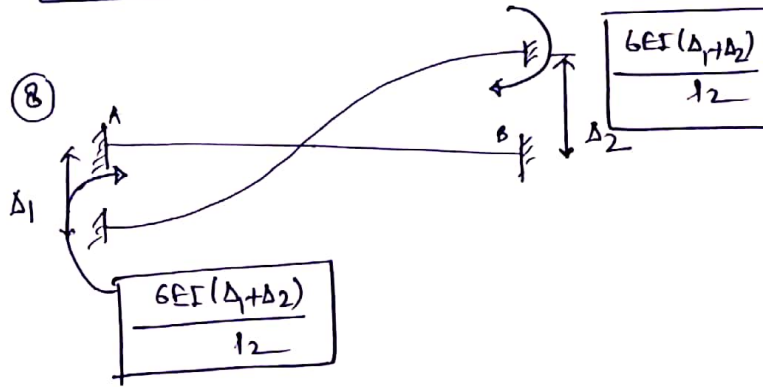
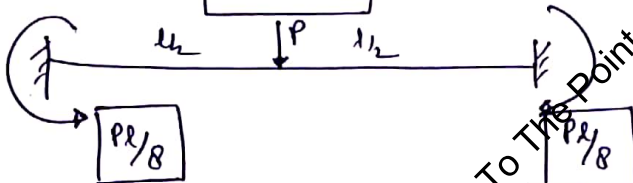
Special case :-

$a = l/2$



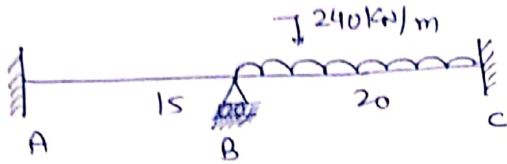
Special case

$a = b = l/2$



moment distribution method type :-
(mostly asked in Exam)

Type-1 :- when both ends of beam is fixed.



$$DF_{BA} = \frac{4EI/15}{4EI/15 + \frac{4EI}{20}} = 0.40$$

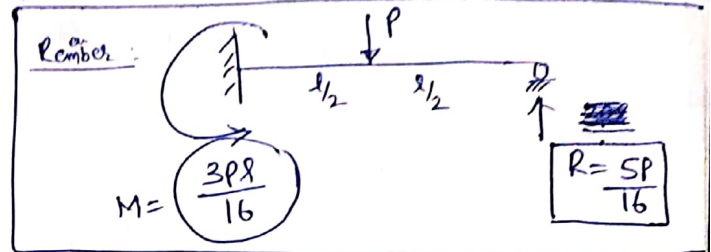
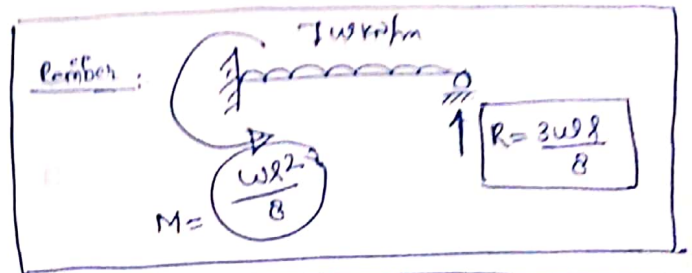
$$\therefore DF_{BC} = 1 - DF_{BA} = 0.60$$

Fixed end moment (anticlockwise = -ve)

$$m_{fbc} = -\frac{wl^2}{12} = -8000 \text{ kNm}$$

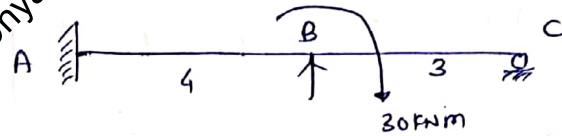
$$m_{fcb} = +\frac{wl^2}{12} = +8000 \text{ kNm}$$

A	B	C
DF ~	0.40	0.60
FEM 0	0	+8000
Balance	+3200	+4800
com +600	-	+2400
Balance -	-	-
end moment	+1600	+10400



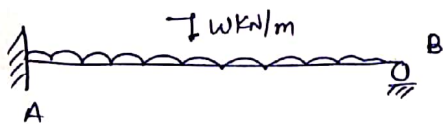
Type-3

when there is concentrated moment between the beam.



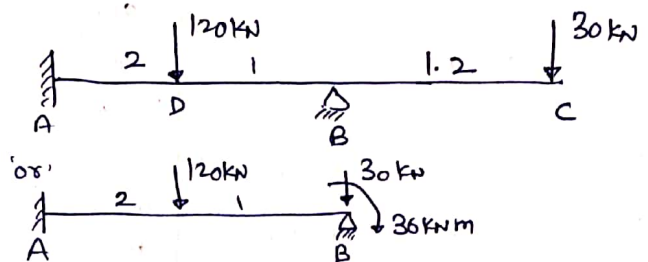
A	B	C
DF ~	2/3	1/3
FEM 0	0	0
Balance	+30x2/3	+30x1/3
com 10	-	-
bal. ~	-	-
+10	+20	+10
	check net +30 kNm	

Type-2 when one end is roller/hinge.



$$M_{fAB} = -\frac{wl^2}{12} \quad m_{fBA} = +\frac{wl^2}{12}$$

A	B
DF -	
FEM $-\frac{wl^2}{12}$	$+\frac{wl^2}{12}$
$-\frac{wl^2}{24}$	$-\frac{wl^2}{12}$
half transfer	



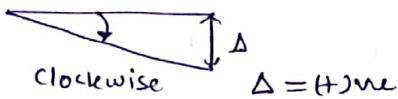
A	B
DF -	
FEM -26.67	+53.33
-17.33	-17.33
half transfer	
-35.33	+36
final fix end moment	

Slope deflection method :-

$$m_{AB} = m_{FAB} + \frac{2EI}{l} [2\theta_A + \theta_B] - \frac{6EI\Delta}{l^2}$$

$$m_{BA} = m_{FBA} + \frac{2EI}{l} [2\theta_B + \theta_A] - \frac{6EI\Delta}{l^2}$$

note:-



modified Slope deflection equation

↳ only use when far end is hinge/roller.

$$m_{AB} = m_{FAB}^* + \frac{3EI}{l} [\theta_A - \frac{\Delta}{l}]$$

$m_{FAB}^* \rightarrow$ end moment
(not a fixed end moment)



$$m_{FAB} = -\frac{wl^2}{12}$$

$$m_{FAB}^* = -\frac{wl^2}{8}$$

unit load method

$$\Delta = \int \frac{M \cdot m}{EI} dx$$

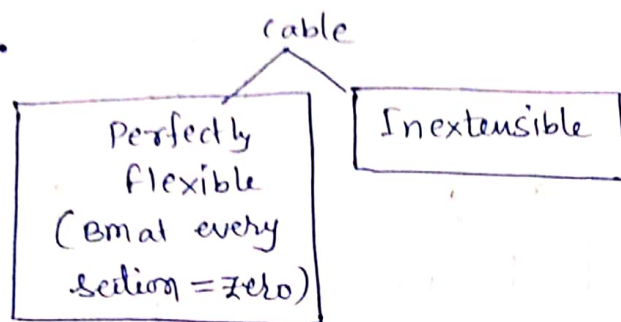
$M \rightarrow$ due to external load

$m \rightarrow$ due to unit load

apply
unit load
in the
direction
in which
deflection
is needed

cable

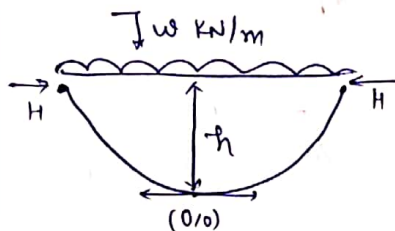
- Cable shape due to self weight $\Rightarrow$ catenary



- cable
 - min. Tension $\Rightarrow$ where sag max.

$T_{min} = H$
 - max. tension = $\sqrt{V_{max}^2 + H^2}$
[at support]
where V_{max} exist $\left\{ \begin{array}{l} V_{max} \rightarrow \text{vertical reaction} \end{array} \right.$

- Symmetrical cable subjected to udl on horizontal span $\rightarrow$ profile parabolic



$$H = \frac{w l^2}{8h}$$

$$y = \left(\frac{w}{2H} \right) x^2$$

$$\therefore \frac{dy}{dx} = \frac{8hx}{12}$$

$$\therefore T = H \sqrt{1 + \left(\frac{dy}{dx} \right)^2} = H \sqrt{1 + \left(\frac{8hx}{12} \right)^2}$$

min
 $x = 0$
↳ max. sag
 $T_{min} = H$

max.
 $x = l/2$ (supports)

$$\text{Cable length } (s) = \int_0^l \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx$$

- when supports are not at same level

$$S = (l_1 + l_2) + \frac{2}{3} \frac{h_1^2}{l_1} + \frac{2}{3} \frac{h_2^2}{l_2}$$

- when support at same level

$$S = l + \frac{8}{3} \frac{h^2}{l}$$

varieties in symmetric parabolic cable of problems

- (i) elastic stretch

$$ds = \frac{HL}{AE} \left(1 + \frac{16}{3} \frac{h^2}{l^2} \right)$$

- (ii) temp. change

$$ds = s \alpha \Delta T = \left(l + \frac{8}{3} \frac{h^2}{l} \right) \alpha \Delta T$$

- (iii) change in sag (dh) due to change in length (ds)

$$\therefore s = l + \frac{8}{3} \frac{h^2}{l} \quad \therefore h^2 = \frac{3l}{8} (s - l)$$

$$\therefore 2h \frac{dh}{ds} = \frac{3l}{8}$$

$$\frac{dh}{ds} = \frac{3l}{16h}$$

- (iv) change in max. tension (dT_{max}) due to change in sag (dh)

$$\therefore T_{max}^2 = V_{max}^2 + H^2$$

$\downarrow$
 $\left(\frac{wl}{2} \right)$

$$2 T_{max} \frac{dT_{max}}{dH} = 2H$$

$$\frac{dT_{max}}{dH} = \frac{H}{T_{max}}$$

- (v) change in Horizontal rxn (dH) due to change in sag (dh)

$$\therefore H = \frac{wl^2}{8h}$$

$$\frac{dH}{dh} = -\frac{H}{h}$$

- (vi) change in sag (dh) due to support slip (dl)

$$\therefore s = l + \frac{8}{3} \frac{h^2}{l}$$

$$\therefore h = \frac{3l}{8} (s - l) \Rightarrow \frac{dh}{dl} = \frac{3}{8} (s - 2l)$$

$$\therefore dh = \frac{3l}{16h} \left(1 - \frac{8}{3} \frac{h^2}{l^2} \right) (-dl)$$

(vii) % reduction in max. Tension = $\frac{dT_{max}}{T_{max}} \times 100$

$$\left(\frac{dy}{dx} = \frac{8hx}{12} \right) \left(x = \frac{l}{2} \right) H \sqrt{1 + \left(\frac{dy}{dx} \right)^2}$$

flexibility & stiffness matrix :-

① f_{ij} → due to unit load at j ,
displacement in the
direction of coordinate.
unit

② k_{ij} → force required/developed
at i for unit displacement at j
unit

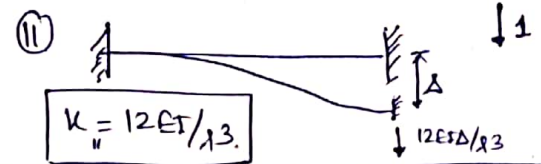
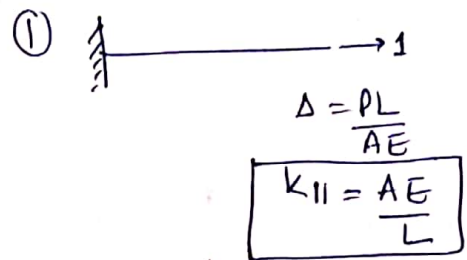
③ $\begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix}^T = \begin{bmatrix} k_{11} & k_{21} \\ k_{12} & k_{22} \end{bmatrix}$ & $\begin{bmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{bmatrix}^T = \begin{bmatrix} f_{11} & f_{21} \\ f_{12} & f_{22} \end{bmatrix}$

[Explained by Maxwell Theorem reciprocal]

④ diagonal terms of matrix are always +ve

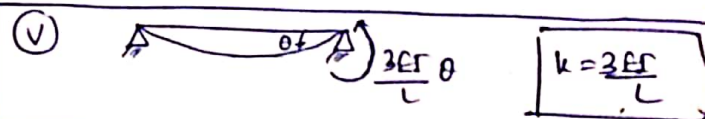
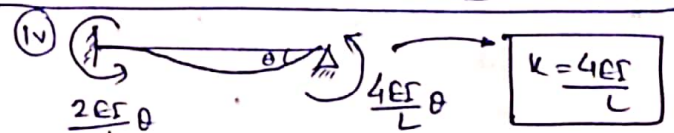
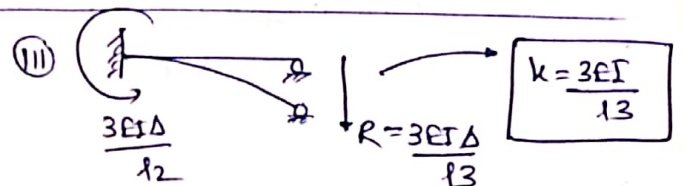
⑤ $CAB KA = CBA KB$

Example :-



$\begin{bmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{bmatrix}^T = \begin{bmatrix} f_{11} & f_{21} \\ f_{12} & f_{22} \end{bmatrix}$

always +ve



flexibility matrix :-

- ① square symmetric matrix.
- ② order of matrix = D_s
($n \times n$)
- ③ each flexibility matrix element = displacement
- ④ diagonal elements = always (+ve)
{ " $\neq 0$, $\neq +ve$ }
- ⑤ valid for stable structure

Stiffness matrix → same property
except order ($n \times n$) = D_k
↓
Degree of kinematic indeterminacy